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St. Joseph's College of Arts & Science (Autonomous)

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B. Sc DEGREE EXAMINATION-APRIL 2021

BC612-BIO TECHNOLOGY

Time: 3 hrs **Max Marks: 75**

SECTION-A (25X1=25)

Answer ALL the questions:

I. Choose the best answer (10X1=10)

1. The first engineered plasmid vector is

   a) pBR 322 b) pUC vectors c) pSC 101 d) pUC 19

2. The most popular and widely used engineered plasmid vector is

   a) pBR 322 b) pUC vectors c) pSC 101 d) pUC 19

3. Disaggregating of cells can be achieved by

   a) physical disruption b) enzymatic digestion

   c) treating with chelating agents d) all of the above

4. Excess CO_2 suppress cell growth and productivity by

   a) inhibiting respiration

   b) altering intracellular pH by diffusing across cell membrane

   c) both a and b

   d) altering pH of the medium

5. The first drug produced using recombinant DNA technology was

   a) streptokinase. b) tPA

   c) insulin. d) penicillin

6. A vaccine can be

   a) an antigenic protein b) weakened pathogen

   c) live attenuated pathogen d) all of these

7. The production of secondary metabolites require the use of

a) protoplast b) cell suspension c) meristem d) auxillary buds

8. The most widely used chemical for protoplast fusion, as fusogens, is

a) Manitol b) Sorbitol c) Mannol d) Poly ethylene glycol (PEG)

9. When energy depleted elements associated with a proton are accepted by an organic molecule, the process is called

a) Fermentation b) Respiration c) Anabolism d) Catabolism

10. Applications of fermentation includes

a) cereal products b) dairy products c) beverage products d) all of above

II. Say true or false (5X1=5)

11. Type II restriction enzymes form homodimers.
12. MEM is the artificial media.
13. Transgenic sheep producing milk contains human blood clotting factor X.
14. Polyamines are not used in plant tissue culture medium.
15. The aeration rate of bioreactor is in the range of 0.25-1.25 vvm.

III. Answer in one sentence (10X1=10)

16. Name any two restriction enzymes.
17. What are cosmids?
18. Define: Sterilization.
19. What is trypsinization?
19. What is subunit vaccine?
20. Name any two traditional vaccines.
21. Define: Plant regeneration.

22. What is somatic embryogenesis?
23. Define: Fermentation.
24. What is freezing?

SECTION-B (4X5=20)

Answer any FOUR of the following:

25. Write note on restriction enzymes.
26. Give the uses of serum free media in animal cell culture.
27. Write the applications of monoclonal antibodies.
28. Give an account on micropropagation.
29. What are the common features of conventional bioreactor?
30. Write note on secondary metabolites production.

SECTION-C (3X10=30)

Answer Any THREE of the following:

31. Write the definition and explain the history and scope of biotechnology.
32. Describe primary cell culture in detail.
33. Write the applications of transgenic animals.
34. How will you isolate the protoplast from plant tissue?
35. Explain the Type 1-fermentation process in detail.

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